AI-generated · Unofficial practice

Chemistry Mock

Paper 1 (Section A Multiple-choice · Section B Structured)

150 minutes（2½ hours）

FOR REVISION

For revision and self-assessment only. Not an official examination paper.

Difficulty rating：**4.0** / 5.0

INSTRUCTIONS

1. This paper consists of Section A (multiple-choice: 24 questions in Part I and 12 in Part II) and Section B (structured questions, Parts I and II).
2. Answer ALL questions.
3. Each question in Section A has only ONE correct answer and carries 1 mark.
4. In Section B, show all working clearly. Unless otherwise specified, give numerical answers to 3 significant figures.
5. For questions marked with an asterisk (＊), 1 mark is awarded for effective communication; answer in complete sentences.
6. A Periodic Table is printed on the last page of this paper.
7. The diagrams in this paper are not necessarily drawn to scale.

＊*Independent, unofficial mock paper. Not affiliated with any official examination authority.*

Section A Multiple-choice Questions

PART I

1. Which of the following pairs of elements, represented by their atomic numbers, belong to the same group of the Periodic Table?

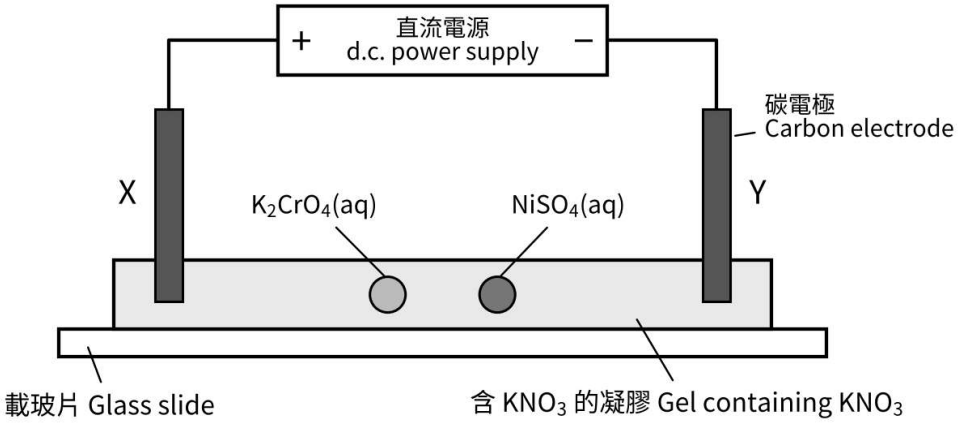
A. 2 and 12

B. 7 and 17

C. 13 and 21

D. 15 and 33

2. As shown in the diagram below, a piece of gel containing potassium nitrate is placed on a glass slide. Carbon electrodes X and Y are inserted at its two ends and connected to a d.c. power supply. Near the centre of the gel, a little apart, a drop of nickel(II) sulphate solution (nearer Y) and a drop of potassium chromate solution (nearer X) are placed separately, and the power supply is switched on for some time. The experiment is then repeated using a gel containing sucrose instead of potassium nitrate. Which of the following correctly describes the expected observations?



A. The green patch moves towards Y and the yellow patch moves towards X; with the sucrose gel, both patches remain essentially unmoved.

B. The green patch moves towards X and the yellow patch moves towards Y; with the sucrose gel, both patches remain essentially unmoved.

C. The green patch moves towards Y and the yellow patch moves towards X; with the sucrose gel, both patches move in the same way.

D. The green patch moves towards X and the yellow patch moves towards Y; with the sucrose gel, both patches move in the same way.

3. Which of the following daily-life processes does NOT involve a redox reaction?

A. Removing stains on clothes with bleach

B. Applying vitamin C on a cut apple to slow down browning

C. Taking an antacid containing magnesium hydroxide to relieve excess stomach acid

D. Silverware turning black in air containing hydrogen sulphide

4. A solid acid $\mathbf{H_nZ}$ has a relative molecular mass of 192.0. 1.60 g of the acid is dissolved in water and made up to $\mathbf{250.0\ cm^3}$ of solution. $\mathbf{20.83\ cm^3}$ of $\mathbf{0.120\ M}$ sodium hydroxide solution is required to completely neutralise $\mathbf{25.00\ cm^3}$ of this solution. What is the value of n?

A. 1 **B.** 2

C. 3 **D.** 4

5. $\mathbf{20.0\ cm^3}$ of $\mathbf{0.30\ M}$ zinc nitrate solution is mixed with $\mathbf{30.0\ cm^3}$ of $\mathbf{0.20\ M}$ sodium hydroxide solution. Assume that zinc hydroxide is completely insoluble in water and that the total volume after mixing is $\mathbf{50.0\ cm^3}$. What is the concentration of $\mathbf{Zn^{2+}(aq)}$ in the resulting solution?

A. 0.060 M **B.** 0.12 M

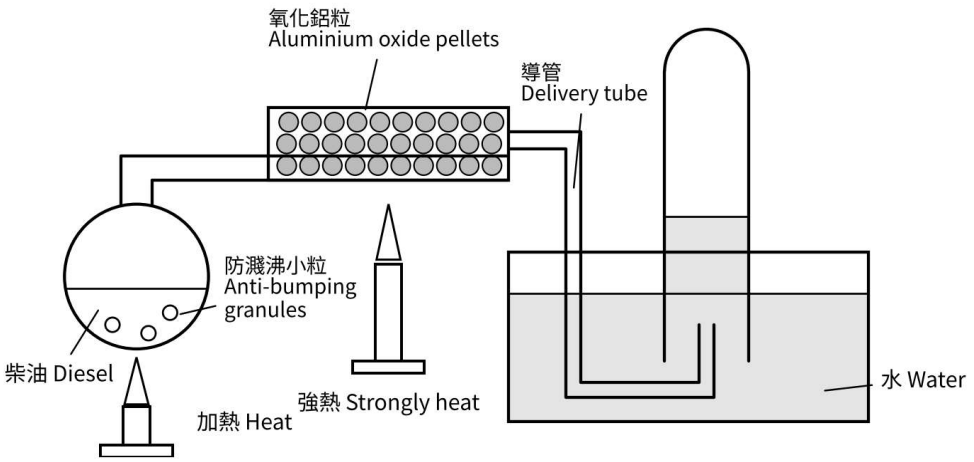
C. 0.15 M **D.** 0.30 M

6. 5.10 g of silver nitrate is strongly heated until there is no further change in mass. 3.24 g of solid residue remains. Which of the following equations can represent the reaction? (Relative atomic masses: N = 14.0, O = 16.0, Ag = 108.0)

A. $\mathbf{AgNO_3(s) \rightarrow AgO(s) + NO_2(g)}$ **B.** $\mathbf{4AgNO_3(s) \rightarrow 2Ag_2O(s) + 4NO_2(g) + O_2(g)}$

C. $\mathbf{2AgNO_3(s) \rightarrow 2AgNO_2(s) + O_2(g)}$ **D.** $\mathbf{2AgNO_3(s) \rightarrow 2Ag(s) + 2NO_2(g) + O_2(g)}$

7. The set-up shown below is used to crack diesel. Which of the following statements concerning this experiment is INCORRECT?



A. The aluminium oxide pellets act as a catalyst so that cracking can occur at a lower temperature.

B. The first test tube of gas collected should be discarded as it mainly contains the air originally present in the set-up.

C. The gas collected can change acidified potassium permanganate solution from purple to colourless.

D. At the end of the experiment, the Bunsen burners should be removed before taking the delivery tube out of the water.

8. 6.85 g of an oxide of lead is heated in a stream of hydrogen until there is no further change in mass. 6.21 g of lead is obtained. What is the empirical formula of this oxide? (Relative atomic masses: O = 16.0, Pb = 207.0)

A. $\mathbf{PbO}$ **B.** $\mathbf{Pb_3O_4}$

C. $\mathbf{Pb_2O_3}$ **D.** $\mathbf{PbO_2}$

9. The following enthalpy changes are given:

$\mathbf{2CH_3COCH_3(l) + 8O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)} \quad \mathbf{\Delta H = -3\,580\ kJ\ mol^{-1}}$

$\mathbf{C(s) + O_2(g) \rightarrow CO_2(g)} \quad \mathbf{\Delta H = -393.5\ kJ\ mol^{-1}}$

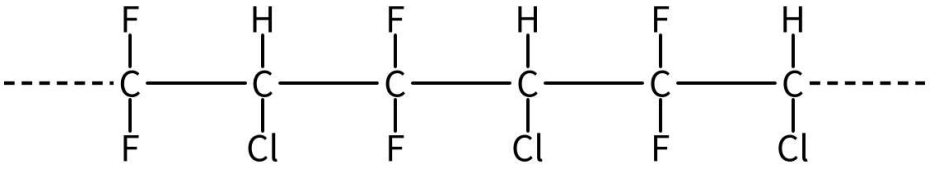
$\mathbf{2H_2(g) + O_2(g) \rightarrow 2H_2O(l)} \quad \mathbf{\Delta H = -571.6\ kJ\ mol^{-1}}$

What is the standard enthalpy change of formation of propanone, $\mathbf{CH_3COCH_3(l)}$?

A. $\mathbf{-1\,105.3\ kJ\ mol^{-1}}$ **B.** $\mathbf{-247.9\ kJ\ mol^{-1}}$

C. $\mathbf{+247.9\ kJ\ mol^{-1}}$ **D.** $\mathbf{+1\,542.1\ kJ\ mol^{-1}}$

10. The diagram below shows part of the chain of an addition polymer. What is the monomer of this polymer?



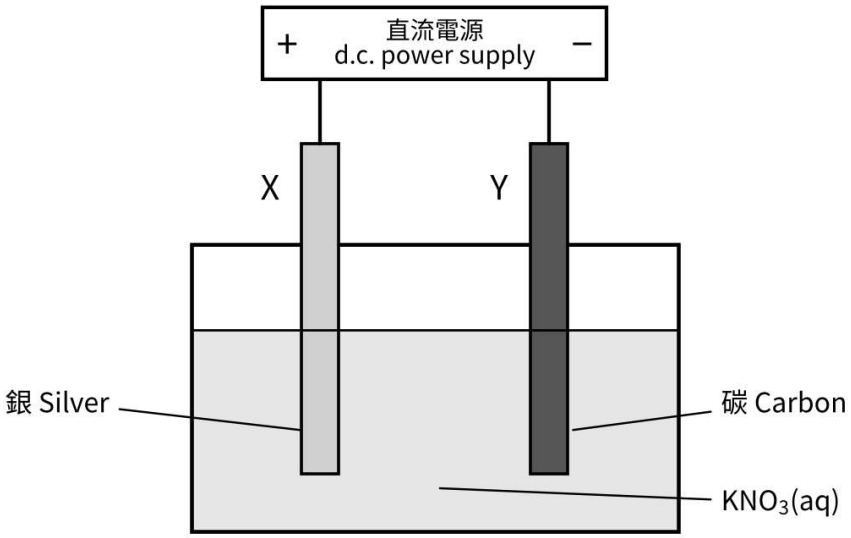
A. $CF_2{=}CHCl$

B. $CHF{=}CFCl$

C. $CHCl{=}CHCl$

D. $CF_2{=}CCl_2$

11. In the electrolysis set-up shown below, electrode X is silver, electrode Y is carbon and the electrolyte is potassium nitrate solution. What are the main reactions at the two electrodes at the beginning of the electrolysis?



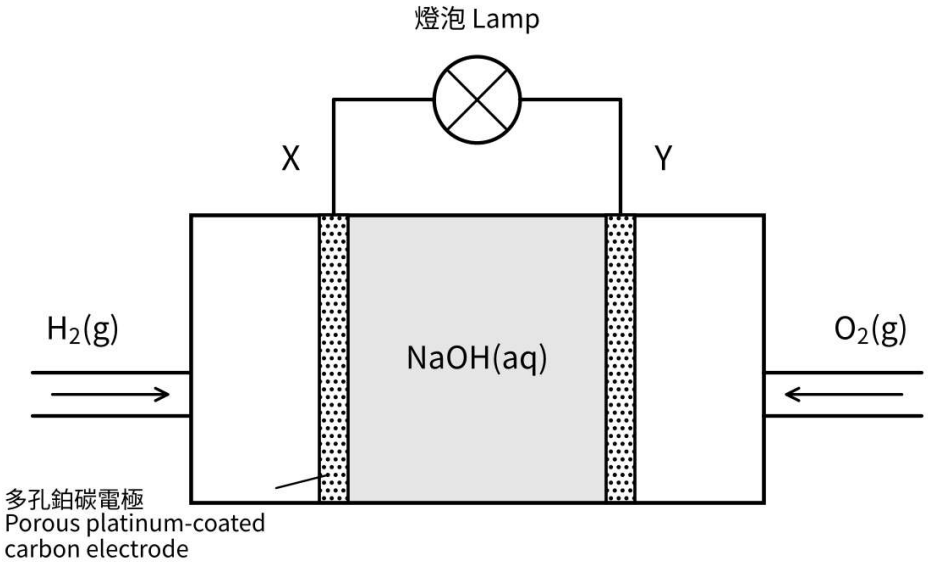
A. X: $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$; Y: $2H^+(aq) + 2e^- \rightarrow H_2(g)$

B. X: $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$; Y: $Ag^+(aq) + e^- \rightarrow Ag(s)$

C. X: $Ag(s) \rightarrow Ag^+(aq) + e^-$; Y: $K^+(aq) + e^- \rightarrow K(s)$

D. X: $Ag(s) \rightarrow Ag^+(aq) + e^-$; Y: $2H^+(aq) + 2e^- \rightarrow H_2(g)$

12. The diagram below shows a hydrogen-oxygen fuel cell using sodium hydroxide solution as the electrolyte. Which of the following statements concerning the operation of this cell is correct?



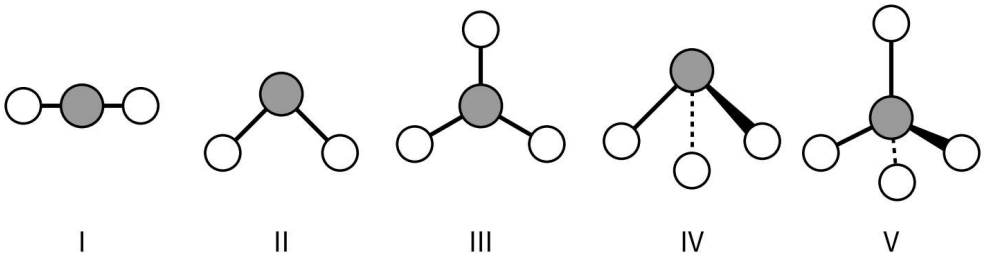
A. The half equation at electrode Y is $\mathbf{O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)}$.

B. The concentration of the sodium hydroxide solution gradually increases.

C. $\mathbf{OH^-(aq)}$ ions in the electrolyte migrate towards electrode X.

D. The cell is a secondary cell because the water formed can be electrolysed.

13. The diagram below shows five molecular shapes I to V (the grey ball represents the central atom). Which of the following combinations concerning molecular shape and polarity is correct?



A. $\mathbf{CH_2Cl_2}$: shape V; polar

B. $\mathbf{BCl_3}$: shape III; polar

C. $\mathbf{PH_3}$: shape III; non-polar

D. $\mathbf{SCl_2}$: shape I; non-polar

14. Consider the following reaction:

$2KMnO_4(s) \rightarrow K_2MnO_4(s) + MnO_2(s) + O_2(g)$

Which of the following statements concerning this reaction is / are correct?

(1) The oxidation number of manganese changes from +7 to +6 and +4 respectively.

(2) $KMnO_4$ acts as both an oxidising agent and a reducing agent.

(3) 2 mol of electrons are transferred for every 1 mol of $O_2(g)$ formed.

A. (1) and (2) only **B.** (1) and (3) only

C. (2) and (3) only **D.** (1), (2) and (3)

15. The steel hull of a cargo ship is immersed in sea water for a long time. Which of the following measures can slow down the rusting of the hull?

(1) attaching magnesium blocks to the hull

(2) attaching nickel plates to the hull

(3) building the hull with stainless steel containing chromium

A. (1) and (2) only **B.** (1) and (3) only

C. (2) and (3) only **D.** (1), (2) and (3)

16. Which of the following tests can be used to distinguish between **1 M** sulphuric acid and **1 M** hydrochloric acid?

(1) adding lead(II) nitrate solution to each acid

(2) adding barium chloride solution to each acid

(3) adding excess limestone chips to each acid

A. (1) and (2) only **B.** (1) and (3) only

C. (2) and (3) only **D.** (1), (2) and (3)

17. Consider the following information about the oxides of metals R, S and T:

Oxide of R: gives metal R when strongly heated with carbon

Oxide of S: no reaction when strongly heated with carbon

Oxide of T: decomposes to metal T on heating alone

Which of the following statements concerning R, S and T is / are correct?

(1) S is more reactive than R.

(2) R can displace T when R is placed in a solution of a salt of T.

(3) The decomposition of the oxide of T on heating shows that T is more reactive than R.

A. (1) and (2) only **B.** (1) and (3) only

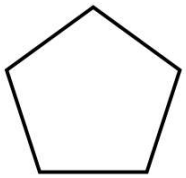
C. (2) and (3) only **D.** (1), (2) and (3)

18. The diagram below shows the structures of compounds X and Y. Which of the following statements concerning X and Y is / are correct?

(1) As X and Y have the same general formula, they belong to the same homologous series.

(2) X and Y are isomers.

(3) Y has cis-trans isomers.



X



Y

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

19. In which of the following reactions does the underlined substance act as a reducing agent?

(1) $Cu^{2+}(aq) + 4\underline{NH_3}(aq) \rightarrow [Cu(NH_3)_4]^{2+}(aq)$

(2) $\underline{H_2O_2}(aq) + 2Fe^{2+}(aq) + 2H^+(aq) \rightarrow 2Fe^{3+}(aq) + 2H_2O(l)$

(3) $\underline{H_2O_2}(aq) + Cl_2(aq) \rightarrow 2H^+(aq) + 2Cl^-(aq) + O_2(g)$

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

20. Which of the following is / are use(s) of calcium hydroxide solution?

(1) treating acidic industrial waste water

(2) distinguishing between $NH_4Cl(s)$ and $NaCl(s)$ by heating with it

(3) drying ammonia gas

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

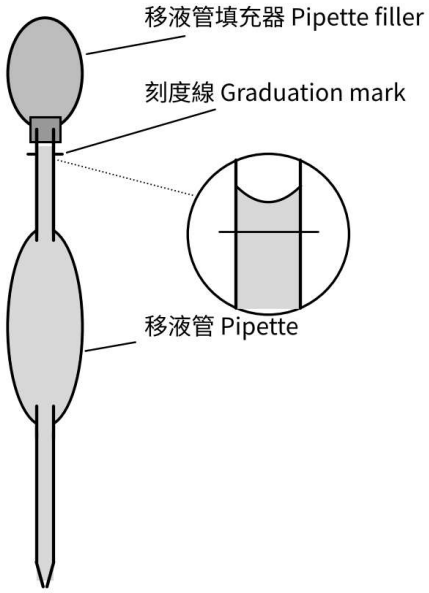
D. (1), (2) and (3)

21. The diagram below shows a student using a pipette to measure **25.00 cm^3** of a sample solution for a back titration (the circle is an enlarged view of the liquid surface during adjustment). Which of the following statements concerning the use of a pipette is / are correct?

(1) The pipette can be used to measure the sample solution immediately after rinsing with distilled water.

(2) With the eye level with the graduation mark, the bottom of the meniscus should touch the mark.

(3) After delivery, the last drop remaining in the tip of the pipette should be blown out.



A. (1) only

B. (2) only

C. (1) and (2) only

D. (2) and (3) only

22. Which of the following is / are exothermic?

(1) thermal decomposition of limestone

(2) condensation of steam to water

(3) dissolving ammonium nitrate in water

A. (1) only

B. (2) only

C. (3) only

D. (2) and (3) only

23. Which of the following statements concerning ethanol is / are correct?

(1) Complete fermentation of 1 mol of glucose gives at most 2 mol of ethanol.

(2) Ethanol contains –OH, so its aqueous solution is alkaline.

(3) When ethanol is heated with acidified potassium dichromate solution, the solution changes from orange to green.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

24. Consider the following statements and choose the best answer:

1st statement: Molten aluminium conducts electricity.

2nd statement: Molten aluminium contains delocalised electrons.

A. Both statements are true and the 2nd statement is a correct explanation of the 1st statement.

B. Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.

C. The 1st statement is false but the 2nd statement is true.

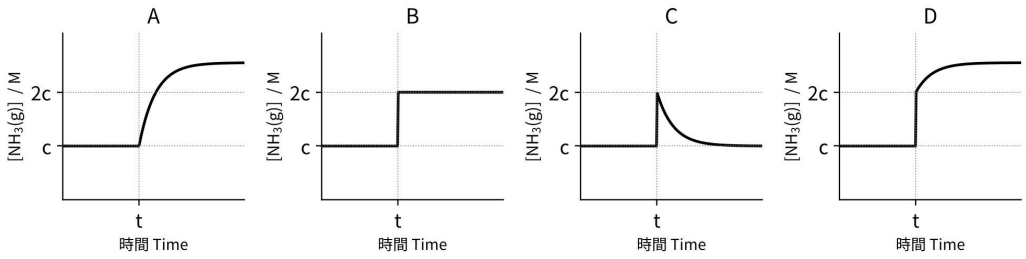
D. Both statements are false.

PART II

25. The following system attains equilibrium at constant temperature in a closed container fitted with a piston:

$\mathbf{N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)}$

The concentration of $\mathbf{NH_3(g)}$ is then $\boldsymbol{c}$. At time $\boldsymbol{t}$, the piston is quickly pushed in so that the volume of the container is halved, while the temperature is kept constant. The system then attains equilibrium again. Which of the following graphs correctly shows the variation in the concentration of $\mathbf{NH_3(g)}$ with time?



A. A

B. B

C. C

D. D

26. At a certain temperature, the equilibrium constant $\boldsymbol{K_c}$ of the reaction $\mathbf{H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)}$ is 64.0. What is the equilibrium constant $\boldsymbol{K_c}$ of the reaction $\mathbf{HBr(g) \rightleftharpoons \frac{1}{2}H_2(g) + \frac{1}{2}Br_2(g)}$ at the same temperature?

A. **0.0156**

B. **0.125**

C. **8.00**

D. **64.0**

27. 0.60 mol of $\mathbf{PCl_5(g)}$ is placed in a closed container of volume $\mathbf{2.0\ dm^3}$. At a certain temperature, the following reaction occurs:

$$\mathbf{PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)}$$

At equilibrium, the container has 0.20 mol of $\mathbf{Cl_2(g)}$. What is K_c of the reaction at this temperature?

A. $\mathbf{0.033\ mol\ dm^{-3}}$ **B.** $\mathbf{0.050\ mol\ dm^{-3}}$

C. $\mathbf{0.067\ mol\ dm^{-3}}$ **D.** $\mathbf{0.10\ mol\ dm^{-3}}$

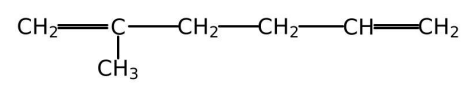
28. At room conditions, a $\mathbf{600\ cm^3}$ gaseous mixture consists of hydrogen and carbon dioxide, with 40% carbon dioxide by volume. The mixture is passed into $\mathbf{0.20\ M}$ sodium hydroxide solution so that all the carbon dioxide is just converted to sodium carbonate, while hydrogen does not react. What is the volume of sodium hydroxide solution required? (Molar volume of gas at room conditions = $\mathbf{24.0\ dm^3}$)

A. $\mathbf{25\ cm^3}$ **B.** $\mathbf{50\ cm^3}$

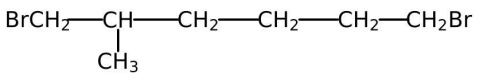
C. $\mathbf{100\ cm^3}$ **D.** $\mathbf{125\ cm^3}$

29. The diagram below shows a compound (the reactant) and four structures I to IV. What is the major organic product when this compound reacts with excess hydrogen bromide?

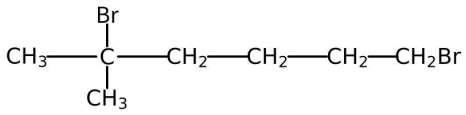
反應物 Reactant



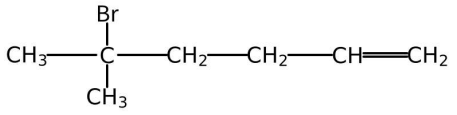
I



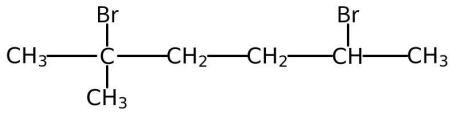
II



III



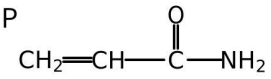
IV

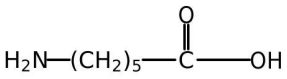


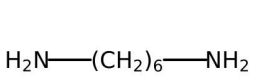
A. I **B.** II

C. III **D.** IV

30. The diagram below shows compounds P, Q and R. Which of the following combinations correctly gives a compound that can undergo addition polymerisation on its own, and a compound that can undergo condensation polymerisation on its own?

P 

Q 

R 

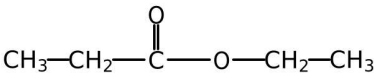
A. addition polymerisation: Q; condensation polymerisation: P

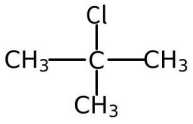
B. addition polymerisation: P; condensation polymerisation: R

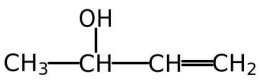
C. addition polymerisation: P; condensation polymerisation: Q

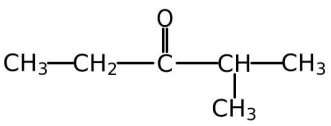
D. addition polymerisation: R; condensation polymerisation: Q

31. The diagram below shows four structures I to IV. Which of the following combinations of structure and systematic name is correct?

I 

II 

III 

IV 

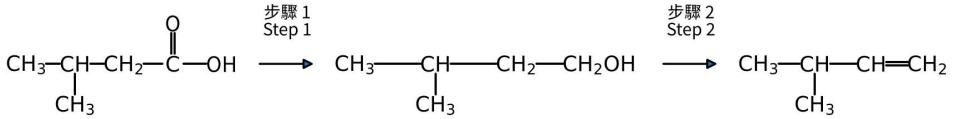
A. I: propyl ethanoate

B. II: 1-chloro-2-methylpropane

C. III: but-1-en-3-ol

D. IV: 2-methylpentan-3-one

32. The diagram below shows a two-step conversion. Which of the following combinations gives suitable reagents and conditions for Step 1 and Step 2?



A. Step 1: acidified $\mathbf{K_2Cr_2O_7(aq)}$, heat; Step 2: conc. $\mathbf{H_2SO_4}$, heat

B. Step 1: $\mathbf{NaBH_4}$; Step 2: conc. $\mathbf{H_2SO_4}$, heat

C. Step 1: $\mathbf{LiAlH_4}$ in dry ether, then acidify; Step 2: $\mathbf{NaOH(aq)}$, heat

D. Step 1: $\mathbf{LiAlH_4}$ in dry ether, then acidify; Step 2: conc. $\mathbf{H_2SO_4}$, heat

33. Which of the following does NOT illustrate a characteristic of manganese as a transition metal?

A. Manganese forms compounds such as $\mathbf{MnCl_2}$ and $\mathbf{KMnO_4}$

B. $\mathbf{MnO_2}$ can speed up the decomposition of hydrogen peroxide

C. $\mathbf{MnO_4^-(aq)}$ is purple

D. Manganese reacts with dilute hydrochloric acid to give hydrogen

34. The following two experiments are carried out to investigate the decomposition of sodium hypochlorite solution catalysed by cobalt oxide powder:

$\mathbf{2NaOCl(aq) \rightarrow 2NaCl(aq) + O_2(g)}$

Experiment I: 25 °C, $\mathbf{100\ cm^3}$ of $\mathbf{0.40\ M}$ NaOCl(aq) with 0.50 g of cobalt oxide

Experiment II: 35 °C, $\mathbf{50\ cm^3}$ of $\mathbf{0.40\ M}$ NaOCl(aq) with 0.50 g of cobalt oxide

Gas volumes are all measured at room conditions. Which of the following statements concerning these two experiments is / are correct?

(1) In comparison with Experiment I, the initial rate in Experiment II is higher.

(2) The final volume of oxygen evolved in Experiment II is half of that in Experiment I.

(3) After the reaction, the mass of cobalt oxide has decreased in both experiments.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

35. The following reaction is carried out in an open beaker at 25 °C:

$$H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow I_2(aq) + 2H_2O(l)$$

Which of the following measurements can be used to follow the progress of this reaction?

(1) the mass of the beaker and its contents

(2) the pH of the reaction mixture

(3) the colour intensity of the reaction mixture

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

36. Consider the following statements and choose the best answer:

1st statement: $(CH_3)_2CHCH(OH)CH(CH_3)_2$ has a pair of enantiomers.

2nd statement: $(CH_3)_2CHCH(OH)CH(CH_3)_2$ is a secondary alcohol.

A. Both statements are true and the 2nd statement is a correct explanation of the 1st statement.

B. Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.

C. The 1st statement is false but the 2nd statement is true.

D. Both statements are false.

END OF SECTION A

Section B Structured Questions

PART I Answer ALL questions.

1. Chlorine and iodine are both in Group VII of the Periodic Table.

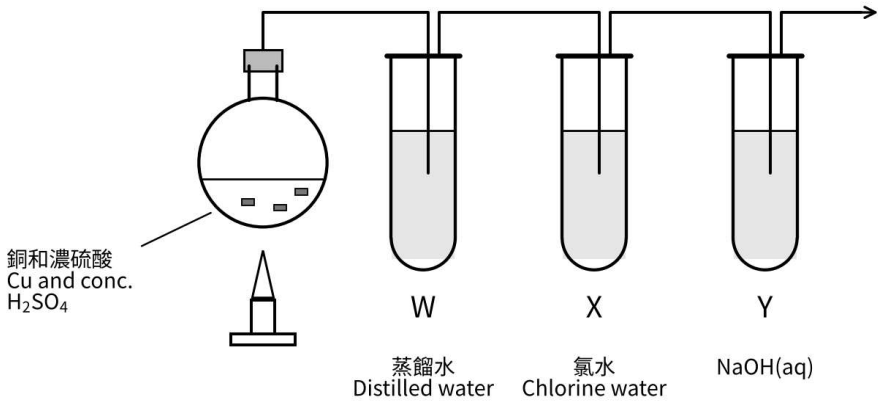
(a) Draw the electron diagram for iodine monochloride, **ICl**, showing electrons in the outermost shells only. (1)

(b) Explain why chlorine is a stronger oxidising agent than iodine. (1)

(c) Sulphur dioxide produced by heating copper with concentrated sulphuric acid (with acid mist removed) is passed through tubes W, X and Y in turn, as shown below. Barium chloride solution and dilute hydrochloric acid are then added to the liquids in tubes W and X separately. The results are as follows:

Tube W: no precipitate

Tube X: white precipitate



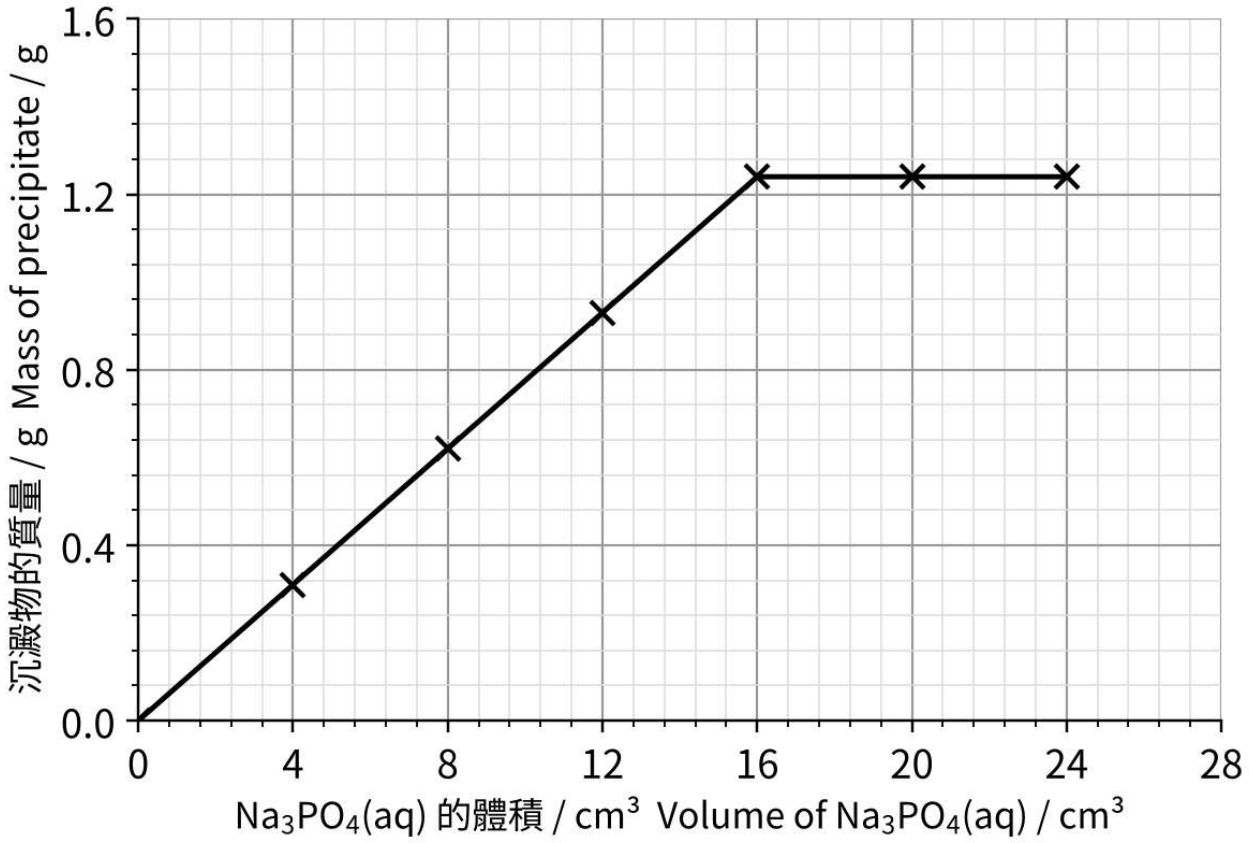
(i) State the expected observation in tube X while sulphur dioxide is being passed in. (1)

(ii) Write an ionic equation for the reaction in tube X. (1)

(iii) With reference to the results of tubes W and X, in terms of oxidation number, explain why sulphur dioxide acts as a reducing agent in tube X. (3)

2. To find the empirical formula of the phosphate of metal M, a student carried out the following experiment:

Different volumes of **0.50 M $Na_3PO_4(aq)$** were added separately to several **40.0 cm^3** portions of **0.30 M $MCl_n(aq)$**. After filtration, each precipitate was washed, dried and weighed. The results are shown in the graph below.



(a) Explain why the mass of precipitate does not increase further after more than **16 cm^3** of **$Na_3PO_4(aq)$** has been added. (1)

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(b)(i) Explain why the precipitate must be washed before weighing. (1)

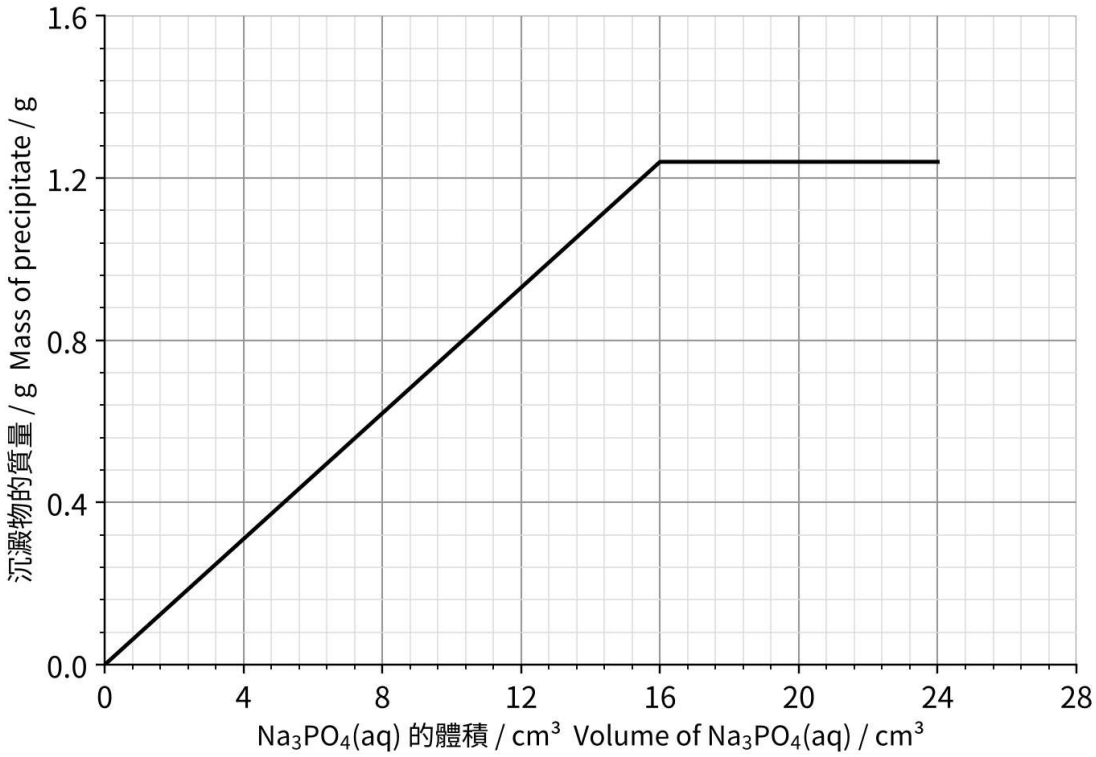
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(ii) The student repeated the experiment using **20.0 cm^3** (instead of **40.0 cm^3**) of the same **$MCl_n(aq)$**. On the graph below, sketch the expected result with a dotted line. (2)



(c) Based on the graph, calculate the empirical formula of this phosphate. (3)

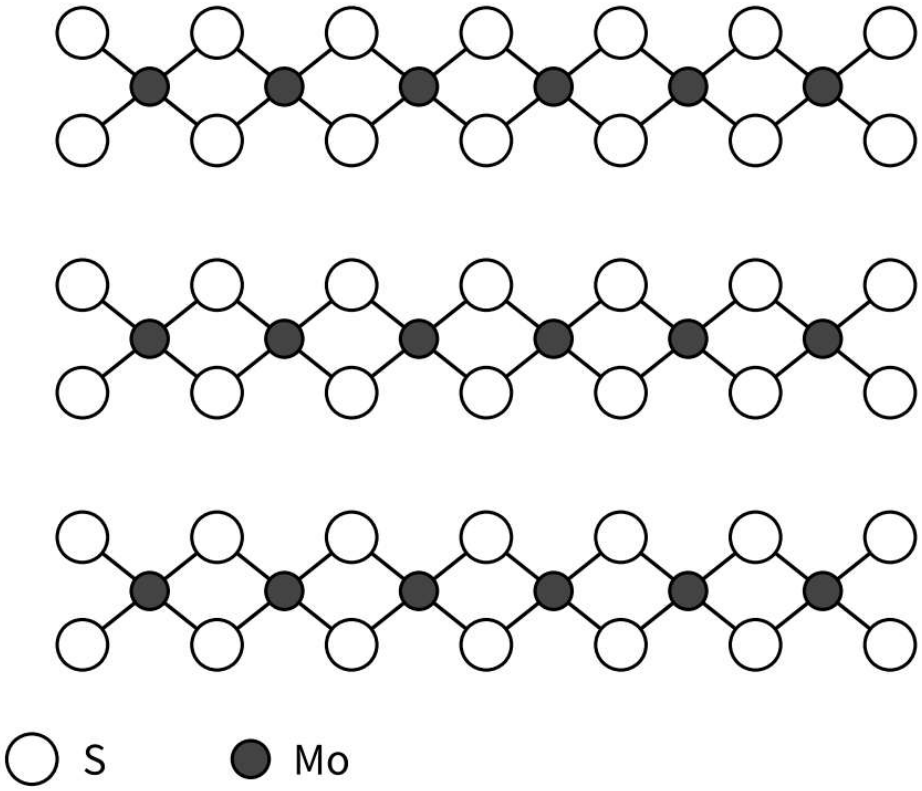
3. The following questions are related to aluminium chloride, phosphorus trichloride and molybdenum disulphide.

(a) Draw the three-dimensional diagrams of a gaseous $AlCl_3$ molecule and a PCl_3 molecule. (2)

(b)(i) At lower temperatures, gaseous aluminium chloride exists as Al_2Cl_6 molecules containing dative covalent bonds. Explain how this type of bond is formed, and draw the structure of Al_2Cl_6, using arrows to represent the dative covalent bonds. (2)

(ii) The boiling point of PCl_3 (76 °C) is much higher than that of PF_3 (−101 °C). Explain this. (2)

(iii) The diagram below shows the layer structure of molybdenum disulphide, MoS_2, which can be used as a solid lubricant. In terms of bonding and structure, explain this use. (2)



4. A sample of quicklime contains calcium oxide and sand, which does not react with acid. To determine the calcium oxide content, 0.500 g of the sample is added to **50.00 cm^3** of **0.300 M** hydrochloric acid (in excess). After the reaction, the mixture in the conical flask is titrated with **0.100 M** sodium hydroxide solution. **18.60 cm^3** of sodium hydroxide solution is needed to reach the end point.

(a) Explain why the quicklime sample is not titrated directly with hydrochloric acid. (1)

(b) Name a suitable indicator for this titration. (1)

(c) Name the apparatus used to measure **50.00 cm^3** of hydrochloric acid accurately. (1)

(d) Before the titration, the mixture in the conical flask is further diluted with distilled water. Would the calculated percentage by mass of calcium oxide be higher than, lower than or the same as the true value? Explain your answer. (1)

(e) Calculate the percentage by mass of calcium oxide in the sample. (Relative atomic masses: O = 16.0, Ca = 40.1) (3)

5. Propanedioic acid is a dibasic acid with the molecular formula $C_3H_4O_4$.

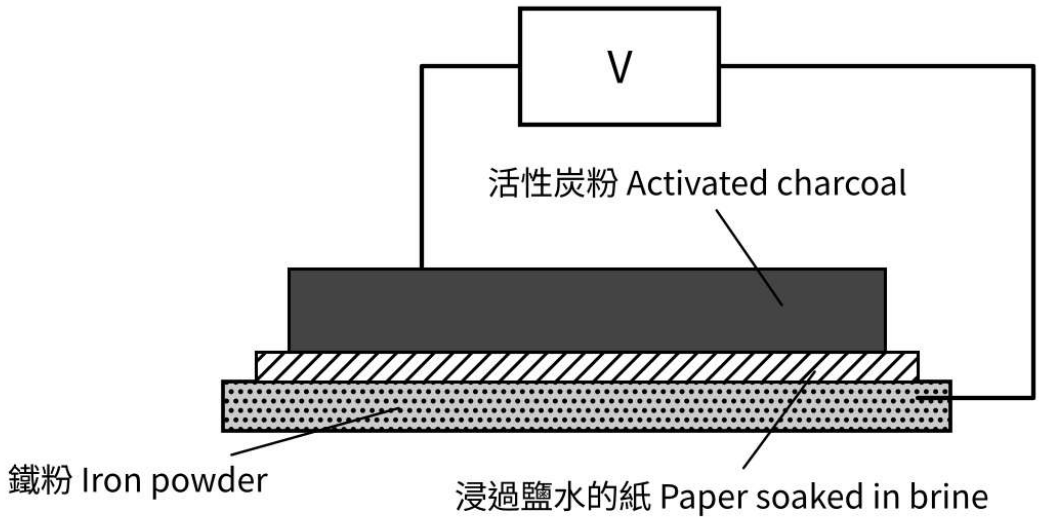
(a) Draw the structure of propanedioic acid. (1)

(b)(i) A propanedioic acid molecule has four hydrogen atoms. Explain why its basicity is 2. (2)

(ii) $25.0\ cm^3$ of $1.00\ M$ propanedioic acid solution and $50.0\ cm^3$ of $1.00\ M$ sodium hydroxide solution are mixed in a polystyrene cup. The temperature rises by 8.30 °C. Assume that the density of the mixture is $1.00\ g\ cm^{-3}$, its specific heat capacity is $4.20\ J\ g^{-1}\ K^{-1}$, and the heat capacity of the cup is negligible. Calculate the enthalpy change of neutralisation for this reaction, in $kJ\ mol^{-1}$ per mole of water formed. (2)

(iii) The enthalpy change of neutralisation of hydrochloric acid and sodium hydroxide solution is $-57.1\ \mathbf{kJ\ mol^{-1}}$. Explain why the result in (b)(ii) is less exothermic. (2)

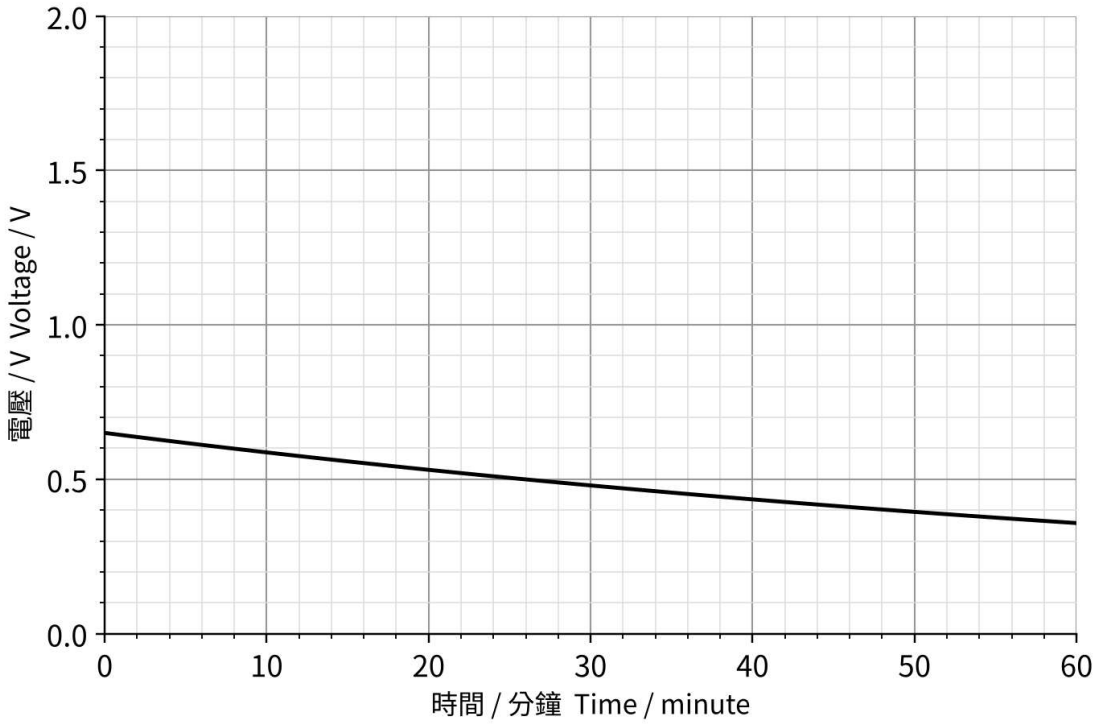
6. Based on the ingredients of a hand warmer, a student made a simple chemical cell using iron powder, paper soaked in brine and activated charcoal powder, as shown below. The voltmeter shows a reading. In this cell, iron is the negative electrode and oxygen in the air is reduced on the surface of the activated charcoal.



(a) Explain why paper soaked in brine, rather than dry paper, is used. (2)

(b) Write the half equations at the iron powder and at the activated charcoal respectively. (2)

(c) The graph below shows the variation in the voltage of this cell with time. If the iron powder is replaced by zinc powder, sketch the expected result on the same graph with a dotted line and give the reason. (1)



(d) After some time, a reddish brown solid appears on the iron powder.

(i) Name this reddish brown solid. (1)

(ii) Suggest one reason why the voltage gradually falls. (1)

7. When hydrated sodium carbonate crystals are mixed with solid ammonium nitrate, the following endothermic reaction occurs:

$$Na_2CO_3 \cdot 10H_2O(s) + 2NH_4NO_3(s) \rightarrow 2NaNO_3(s) + 2NH_3(g) + CO_2(g) + 11H_2O(l)$$

(a) Describe how to show in a school laboratory that this reaction is endothermic. (2)

(b) In view of laboratory safety, suggest one precaution for this experiment and give the reason. (1)

(c) The following standard enthalpy changes of formation ($\mathbf{kJ\ mol^{-1}}$) are given:
$\mathbf{Na_2CO_3 \cdot 10H_2O(s)}$: −4 081; $\mathbf{NH_4NO_3(s)}$: −366; $\mathbf{NaNO_3(s)}$: −468; $\mathbf{NH_3(g)}$: −46.1; $\mathbf{CO_2(g)}$: −393.5; $\mathbf{H_2O(l)}$: −285.8

(i) Write the thermochemical equation representing the standard enthalpy change of formation of $\mathbf{NH_4NO_3(s)}$. (1)

(ii) Calculate the standard enthalpy change of the above reaction, in $\mathbf{kJ\ mol^{-1}}$. (2)

(iii) If the enthalpy change of this reaction is found by measuring the temperature change in a polystyrene cup, suggest one reason why the value obtained is less endothermic than that in (c)(ii). (1)

8. Poly(propene) is a common plastic.

(a) *Outline how poly(propene) can be made from crude oil. Your answer should include the processes and reaction conditions involved, and suitable chemical equation(s). (6)

PART II Answer ALL questions.

9. At 25 °C, $\mathbf{30.0\ cm^3}$ of $\mathbf{0.20\ M}$ silver nitrate solution is mixed with $\mathbf{20.0\ cm^3}$ of $\mathbf{0.20\ M}$ iron(II) nitrate solution to give the following equilibrium system:

$$\mathbf{Ag^+(aq) + Fe^{2+}(aq) \rightleftharpoons Ag(s) + Fe^{3+}(aq)}$$

At equilibrium, the concentration of $\mathbf{Fe^{3+}(aq)}$ is $\mathbf{0.020\ M}$.

(a) Calculate the equilibrium constant $\boldsymbol{K_c}$ of this reaction at 25 °C. (3)

(b) At 35 °C, K_c of this reaction is $2.53\ mol^{-1}\ dm^3$. Deduce whether the forward reaction is exothermic or endothermic. Explain your answer. (2)

10. Propan-2-ol can be made from 2-bromopropane or from propan-2-yl ethanoate.

(a) 2-Bromopropane is heated with sodium hydroxide solution to give propan-2-ol.

(i) State the type of this reaction. (1)

(ii) Write the chemical equation for this reaction. (1)

(b) Propan-2-yl ethanoate, $CH_3COOCH(CH_3)_2$, is heated under reflux with sodium hydroxide solution to give propan-2-ol, the product in (a).

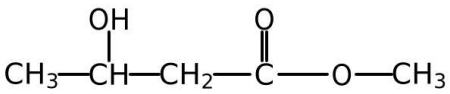
(i) Write the chemical equation for this reaction. (1)

(ii) Draw a labelled diagram of the set-up for heating under reflux. (2)

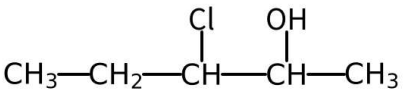
(c) Acrylic fibre is made by the addition polymerisation of acrylonitrile, $CH_2=CHCN$. Draw the repeating unit of polyacrylonitrile. (1)

11. The diagram below shows the structures of compounds E, G, J and L.

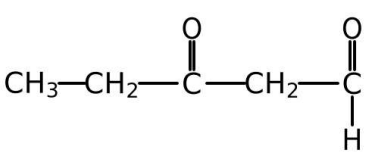
E



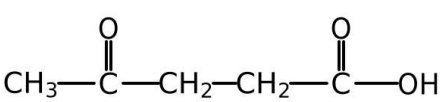
G



J



L



(a) Is the hydroxyl group in E that of a primary, secondary or tertiary alcohol? (1)

(b) How many chiral centres are there in a molecule of G? (1)

(c)(i) J and L are separately warmed with acidified potassium permanganate solution. State the expected observation that distinguishes them. (1)

(ii) Magnesium ribbon is added separately to aqueous solutions of E and L. State the expected observation that distinguishes them. (1)

(d) Outline a synthetic route, with NO MORE THAN THREE STEPS, to convert E to $CH_3CH=CHCOOH$. For each step, give the reagent(s), reaction conditions (as appropriate) and structure of the organic product. (3)

12. The reaction between peroxodisulphate ions and iodide ions is very slow:

$S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$

The reaction becomes much faster after a small amount of $Fe^{2+}(aq)$ is added.

(a) In terms of the characteristics of transition metals, explain how $Fe^{2+}(aq)$ speeds up the reaction, and judge whether $Zn^{2+}(aq)$ could have the same effect. (4)

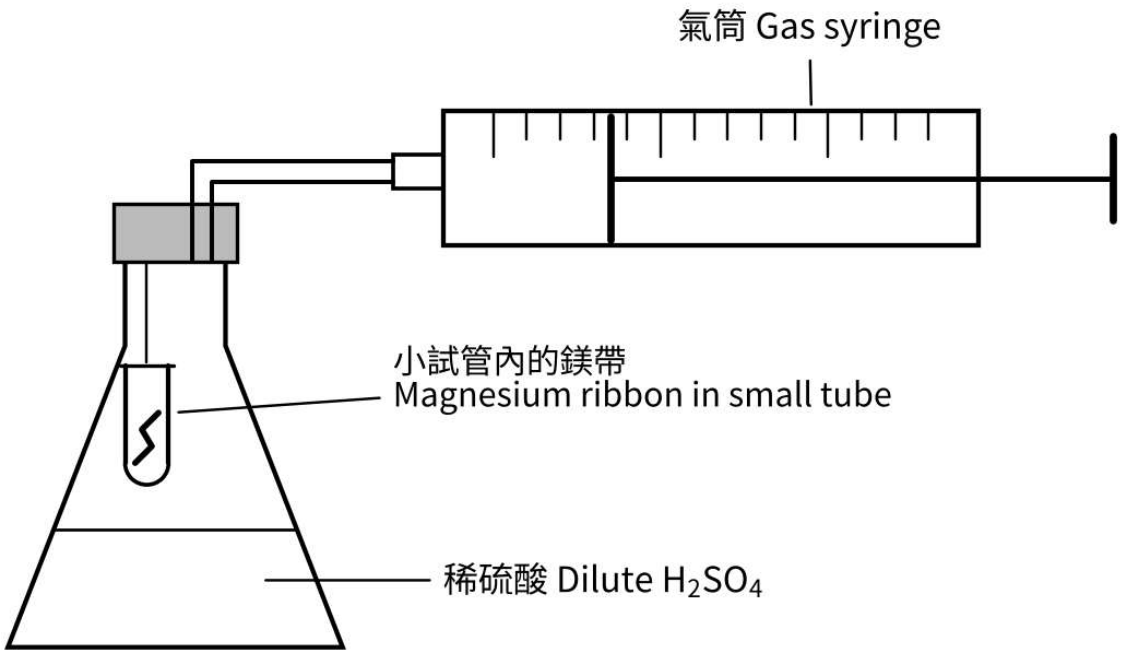
13. You are provided with the following materials and information:

• magnesium ribbon (in excess)

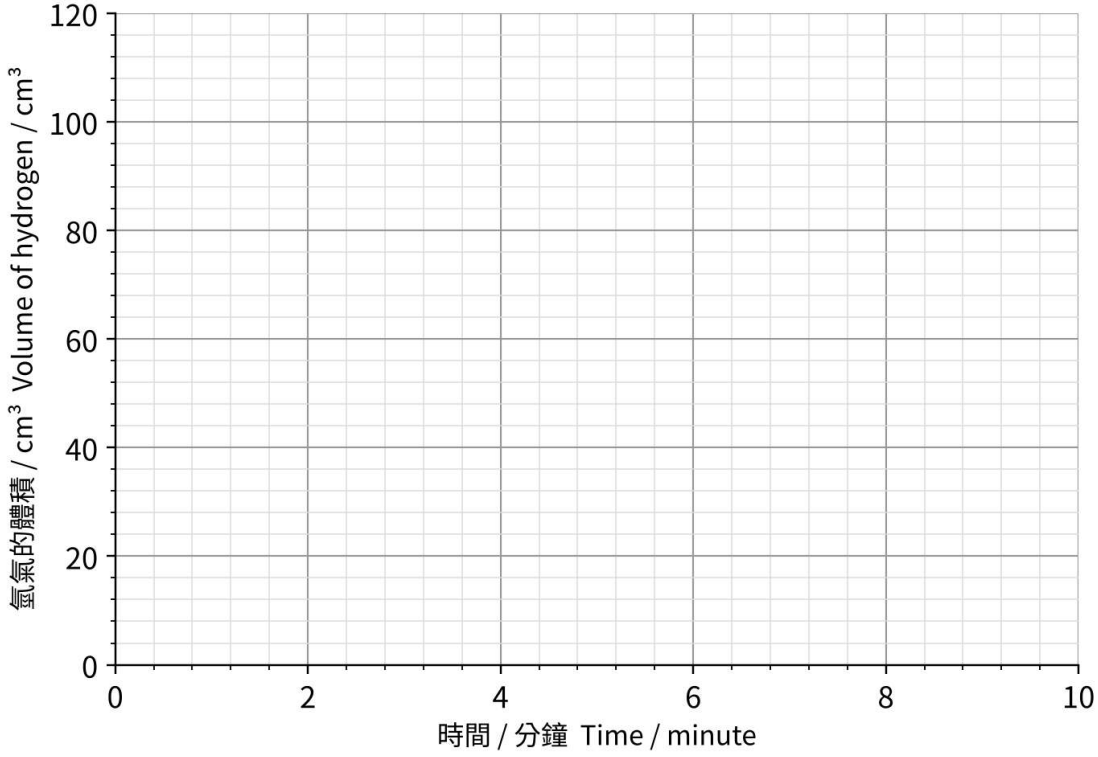
• **0.20 M** and **0.10 M** sulphuric acid

• the set-up shown below, and a stopwatch

(Molar volume of gas at room conditions = **24.0 dm^3**)



(a) *Describe how to use the items provided to investigate the effect of the concentration of sulphuric acid on the rate of this reaction, using **20.0 cm^3** of sulphuric acid in each experiment. On the graph below, sketch the two curves expected, labelling their final volumes, and at the molecular level, explain the differences between the two curves. (6)



END OF PAPER

Periodic Table

I	II											III	IV	V	VI	VII	0
1 H 1.0																	2 He 4.0
3 Li 6.9	4 Be 9.0											5 B 10.8	6 C 12.0	7 N 14.0	8 O 16.0	9 F 19.0	10 Ne 20.2
11 Na 23.0	12 Mg 24.3											13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.5	18 Ar 40.0
19 K 39.1	20 Ca 40.1	21 Sc 45.0	22 Ti 47.9	23 V 50.9	24 Cr 52.0	25 Mn 54.9	26 Fe 55.8	27 Co 58.9	28 Ni 58.7	29 Cu 63.5	30 Zn 65.4	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8
37 Rb 85.5	38 Sr 87.6	39 Y 88.9	40 Zr 91.2	41 Nb 92.9	42 Mo 95.9	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)															

atomic number
Symbol
relative atomic mass

Shaded: transition elements. Only the first lanthanoid and actinoid are shown. Values in brackets are mass numbers of the most stable isotopes.

∗Original layout.